

In-Field Comparison Among Polarization-Mode-Dispersion Measurement Techniques

Andrea Galtarossa, *Member, IEEE*, Giovanni Gianello, Carlo G. Someda, *Senior Member, IEEE*,
and Marco Schiano, *Member, IEEE*

Abstract—We report polarization mode dispersion measurements on a 168 km long undersea optical cable, performed with two techniques. From experimental results, statistical behavior of polarization parameters and their time fluctuations have been extracted. An average differential group delay of about 13 ps shows that dispersion-shifted fibers can be affected by a polarization dispersion significantly larger than standard step-index fibers. The correlation length of the polarization-coupling perturbation in the tested fibers is not negligible with respect to the link length.

I. INTRODUCTION

SINGLE-MODE FIBERS are always birefringent, either intentionally or at random, so that the fundamental $LP_{0,1}$ mode cannot be considered truly double degenerate. The two polarization modes have distinct propagation constants $\beta_j(\omega)$ ($j = 1, 2$), that can be expanded as Taylor series in ω , around the central frequency of the signal band. When the second-order term in the series, β'' , i.e., the chromatic dispersion (CD), is small, the dominant contribution to signal distortion is the polarization mode dispersion (PMD), i.e., the differential group delay (DGD) $\beta'_1 - \beta'_2$ between polarization modes. While several techniques have been proposed to compensate CD, PMD compensation is very difficult, because it is a random process and because DGD fluctuates versus time. PMD has been widely investigated theoretically [1]–[3] and experimentally [4]–[10] in the last years. At present, emphasis appears to be on investigation of penalties affecting long-haul, very high-bit-rate systems, and on comparison and standardization of measurement techniques. Several reports confirmed that standard step-index fibers (zero chromatic dispersion around 1300 nm) are affected by a completely random birefringence, if they are loosely cabled. As a consequence, their DGD is quite low (less than $0.1 \text{ ps}/\sqrt{\text{km}}$, see, for example, [6]). On the other hand, the same fibers may be affected by a deterministic birefringence, superimposed on the random one, when they are embedded in a ribbon [11]. On dispersion-shifted (DS) fibers, the situation is drastically different, as the literature contains just a few, and very controversial, results [4], [7], [12]. In this work, we first present the results of a set of PMD measurements that were performed on DS fibers

which belong to an installed undersea optical cable. Several techniques have been used, so that results can be compared. Next, we draw from this comparison some recommendations about measurement procedures, in particular for what regards accuracy and speed of measurements, which are both crucial in field tests. Our main conclusions are that all methods are suitable for DGD field test, even if Poincaré sphere method may be preferable, because it permits a complete characterization of PMD in a spectral range of arbitrary width.

II. EXPERIMENTS

All the measurements were performed on a pair of DS fibers, belonging to an optical cable installed, between Grosseto and Civitavecchia, as a section of the Italian optical cable network referred to as “Festoons.” The total section length is 168 km, and consists of two terrestrial trunks (18 km on the Northern side, 11 km on the Southern one), and a long undersea trunk (139 km). The other 10 fibers in the cable carry commercial telecommunication traffic. The total attenuation of the link is large (around 37.3 dB). As a consequence, the transmitting and the receiving sections of the PMD measurement set-ups were located one at each end of the link, in Grosseto and in Civitavecchia, respectively. Among the known PMD measurement techniques, we chose to use directly the fixed polarizers method (FP) and the Poincaré sphere method (PS). Data equivalent to the interferometric method (IF) were derived Fourier-transforming the FP data, while data equivalent to the state of polarization method (SOP) were obtained by suitably processing the PS data. The FP and PS techniques were selected aiming at good measuring accuracy, simple transportation of the equipment, and low costs, i.e., the most relevant properties for field tests. An example of low-cost and high-simplicity solution was the choice of the tunable source, for both methods. A temperature-tuned DFB laser was preferred to the more sophisticated external-cavity laser. The set-up schematic diagram is shown in Fig. 1. The DFB laser (source for the PS method) could be temperature-tuned between 1549 and 1553 nm, in steps of 0.1 nm. The broadband source required by the FP method was the strong spontaneous emission of an EDFA (+18 dBm output power). The two detection sections consisted of a fast polarimeter (for PS) and of an optical spectrum analyzer (OSA) (for FP). Measurements with the two methods were performed simultaneously, in order

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A. Galtarossa, G. Gianello, and C. G. Someda are with the Department of Electronics and Informatics, University of Padova, Padova I-35131, Italy.

M. Schiano is with SAIFO srl, Padova I-35127, Italy.

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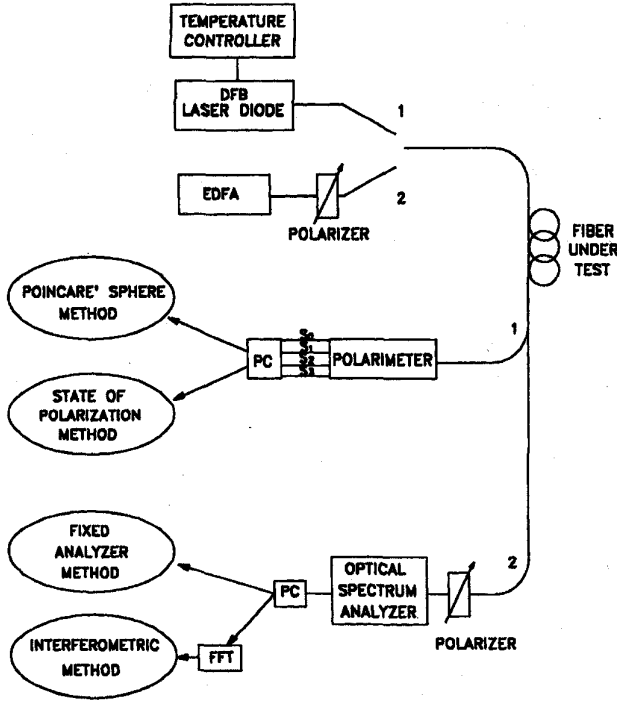


Fig. 1. General scheme of set-up; branch 1 refers to PS and SOP methods, while branch 2 refers to FP and IF methods.

to avoid ambiguities due to fluctuations of DGD versus time. For obvious advantages in paper readability, the results will be presented in sequence. Definitions that can be found in the literature, concerning PMD and its measurements, are not unique and unambiguous. Therefore, before showing the experimental results, we will briefly review the basic equations of each method. The reader who is interested in more details on PMD measurement methods may consult [10], [13]–[15].

III. EXPERIMENTAL RESULTS

A. The Fixed-Polarizer Method

Details on this method can be found in [16]. All the definitions given or used in this Section are in perfect agreement with this reference. The FP method entails us, first of all, to determine whether the fiber under test behaves as predicted by the “long-length regime” ($L/l_c \rightarrow \infty$), or by the “short-length regime” ($L/l_c \rightarrow 0$), L being the fiber length and l_c being the correlation length, i.e., the fiber length over which the polarization of a lightwave becomes uncorrelated with the input polarization, due to polarization-mode coupling. In the short-length regime, the DGD $\Delta\tau$ can be calculated as follows:

$$\Delta\tau = k \frac{\pi N_e}{\Delta\omega} = \frac{\pi N_m}{\Delta\omega}, \frac{\langle N_e \rangle}{\langle N_m \rangle} = 1, L/l_c \rightarrow 0 \quad (1)$$

while in the long-length regime it is given by

$$\begin{aligned} \langle \Delta\tau \rangle &= k \frac{\pi \langle N_e \rangle}{\Delta\omega} = 4 \frac{\langle N_m \rangle}{\Delta\omega}, \\ \frac{\langle N_e \rangle}{\langle N_m \rangle} &= 1.54, L/l_c \rightarrow \infty. \end{aligned} \quad (2)$$

The parameters in these formulas are defined as follows:

$\langle N_m \rangle$ is the expected number of times that the normalized transmission curve crosses the mean value 0.5 in a frequency window of width $\Delta\omega$,

$\langle N_e \rangle$ is the expected number of maxima and minima of the normalized transmission curve in a frequency window of width $\Delta\omega$, and

k is the polarization coupling factor; $k = 1$ in the *short-length regime* (linearly birefringent fiber) and $k = 0.824$ in the *long-length regime* (completely random birefringence).

We are allowed to assume [16] that $N_e \approx \langle N_e \rangle$ and $N_m \approx \langle N_m \rangle$. Also in [16], empirical formulas for the statistical uncertainties (σ^2) of the measurements were also derived

$$\left[\frac{\sigma^2}{\langle \Delta\tau \rangle^2} \right]_{\text{mean-level}} \approx \frac{0.7}{N_m} \quad (3)$$

$$\left[\frac{\sigma^2}{\langle \Delta\tau \rangle^2} \right]_{\text{extrema}} \approx \frac{0.5}{N_e}. \quad (4)$$

Let us stress two comments on these results:

- 1) due to the peculiar shape of the typical normalized transmission curves [16], which do not always intersect the mean value between two consecutive extrema, the statistical uncertainty of a DGD measurement based on the count of extrema is smaller than that of the corresponding measurement based on mean-level crossing measurements by about 30%;
- 2) the DGD can be evaluated using the count of extrema only if the coupling coefficient k is a known quantity.

Usually, standard step-index fibers can be thought of as being in the long-length regime. Then, it is suggested [15], [17] to take $0.81 < k < 0.85$. On the contrary, we will show now that giving to k an a-priori selected value can lead to erroneous results for dispersion-shifted fibers, because in these fibers the ratio L/l_c has a finite value and, as a consequence, DGD can be about one order of magnitude larger than in standard fibers. In fact, k can be determined from experimental data, through the following relation, which can be derived from [16]:

$$k = \frac{4}{\pi} \langle \langle \frac{N_m}{N_e} \rangle \rangle \quad (5)$$

where the symbol $\langle \langle * \rangle \rangle$ means first, that the ratio between N_m/N_e is measured over a very large frequency window (50 nm), which is equivalent to averaging over wavelength; second, that the measurement is repeated several times, at time intervals large enough for the samples to be uncorrelated, and then averaged (time average, or ensemble average, as the process is stationary). In particular, in our experiment, measurements were repeated for 25 h, at time intervals of 30 min. We found $k = 0.916$, which corresponds to a fiber in a “quasilong-length” regime, i.e., to a ratio $L/l_c \gg 1$, but far from infinite. A large frequency window implies, obviously, a time consuming measurement, during which the cable DGD may fluctuate. Therefore, we derived next, experimentally, the minimum wavelength window which is necessary in order

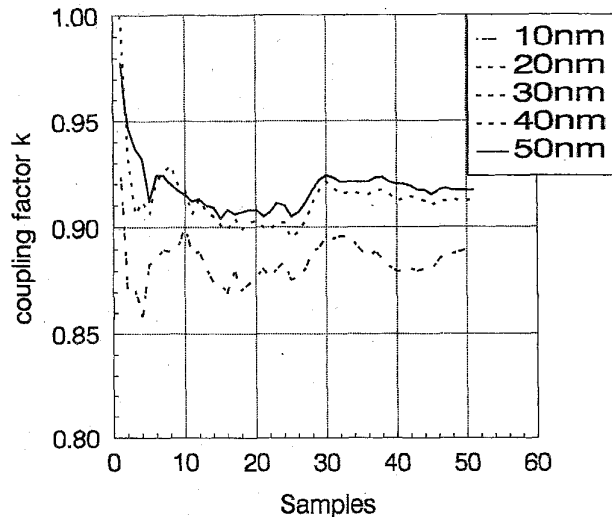


Fig. 2. Coupling factor k as a function of number of samples, calculated for 5 different wavelength widths: 10 nm (dashed-dotted line), 20-30-40 nm (dotted lines), and 50 nm (continuous line).

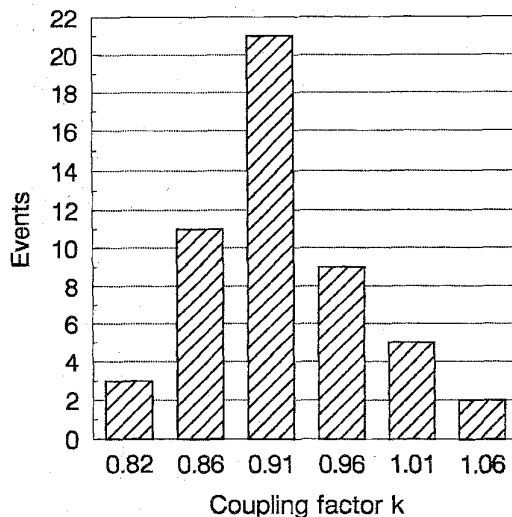


Fig. 3. Histogram showing the distribution of coupling factor in a wavelength width of 50 nm.

to evaluate the coupling factor k with an accuracy better than 1%. The results are illustrated in Fig. 2, where k is plotted versus number of samples, for five different wavelength window widths. It shows that, when $N_e \approx 70$, k reaches its asymptotic value after ≈ 30 samples. The corresponding spectral width is about 20 nm. Larger spectral windows are not necessary, as the accuracy remains practically the same. On the other hand, when $N_e < 40$, it is still possible to determine k with an accuracy of a few percent, but the minimum number of samples and, consequently, the time to perform the measurements, must be enlarged. Fig. 3 shows the k statistical distribution as a function of sample number (thus, as a function of time), for a spectral width of 50 nm.

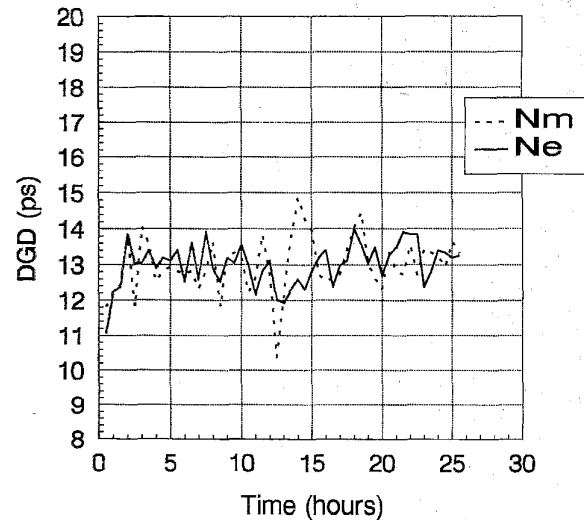


Fig. 4. Time evolution of DGD's calculated using zero-crossing values (dotted line) and extrema values (continuous line) for a spectral width of 50 nm.

The distribution is centered around $k = 0.92$, reconfirming the value given in Fig. 2. Fig. 4 shows the DGD obtained using the definitions based on the number of zero crossings, N_m , and on the number of extrema, N_e as a function of time over 25 h. The two curves have obviously the same average value (13.01 ps), but most of the samples give two different values for the DGD, for the two different definitions. The accuracy of the single DGD value is calculated to be 7% and 5%, respectively, using the mean-cross approach, or the extrema one. They correspond to error bars of less than 1 ps and less than 0.7 ps, respectively. From these results, we may draw the following comments. The FP method has the following drawbacks:

- a small amount of information can be derived on the statistical properties of PMD;
- only large $\langle \Delta\tau \rangle$ can be measured easily, and in any case a very broadband light source is required;
- there is an intrinsic danger of misinterpreting the measurements; in particular, a direct count of either N_e or N_m is not always unambiguous, and therefore polynomial interpolation of data is recommended.

B. The Interferometric Method

As we said in the Introduction, we did not perform any in-field test with this method; but, as well known, results obtained by the IF method are related by a Fourier transform to results obtained by the FP method [10, 14]. DGD can be determined by means of the IF method both in the short-length and in the long-length regime. As known, the fringe visibility, as a function of the interferometer unbalance, exhibits two peaks in the short-length regime, and becomes close to Gaussian in the long-length regime. It has already been pointed out [10] that DGD measurements with the IF method becomes ambiguous when the perturbation correlation length is comparable to the fiber length. One of the purposes of our investigation was to check whether similar ambiguities remain when the IF

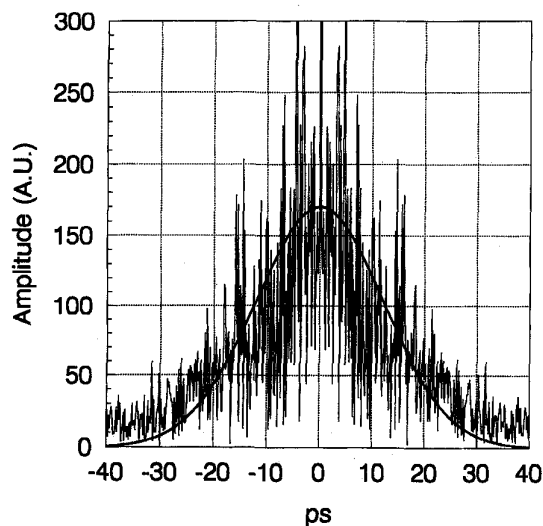


Fig. 5. FFT of output signal and its Gaussian fit.

data are obtained from FP data via the Fourier transform. We recorded the output data from the optical spectrum analyzer while making the FP tests, and then processed them on a PC. Fig. 5 shows an example of results. We see that, in the “quasilong-length” regime, a Gaussian envelope fits well the data. From it, the DGD can be calculated as shown in [14].¹ Fig. 6 illustrates the DGD evolution as a function of time corresponding to the measurements described in the previous paragraph. Theoretically, the curve should be the same as in Fig. 4. In practice, the agreement between the two curves is quite satisfactory, if we account for errors of a few percent on the data, plus signal fitting. The average DGD obtained from the Gaussian fits to the envelopes of the transforms of all our data is 12.57 ps, i.e., it differs about 3% from DGD calculated with the FP method. To conclude, the drawbacks of this indirect IF method are, obviously, the same as those of the FP method, but FFT processing of data can be faster than counting extrema or crossings, and may yield a more accurate DGD determination. On the other hand, this procedure does not provide information about the regime of the fiber. Furthermore, the DGD is averaged over a very large spectral range, so that information about the DGD behavior as a function of wavelength is completely lost.

C. The Poincaré Sphere Method

As is well known, all the information on PMD (the principal states of polarization, their DGD, its time and wavelength dependences) can be derived from the arc described on the Poincaré sphere by the point representing the output state of polarization as a function of wavelength, for a constant input state of polarization. The DGD is then calculated using the following formula

$$\Delta\tau(\omega) = \frac{\Delta\Phi}{\Delta\omega} \quad (6)$$

¹In a paper to be published, B.L. Heffner shows that results obtained using IF and PS methods may be related in a different way than in [14]

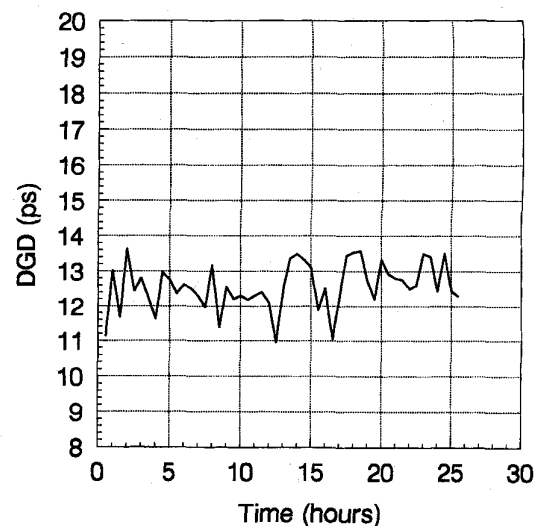


Fig. 6. Time evolution of DGD determined by means of IF method; FFT's have been performed over a window width of 50 nm.

TABLE I

$\langle\langle \Delta\tau \rangle\rangle$	$\langle \sqrt{\sum_i \Delta\tau_i^2} \rangle$	$\langle STD \rangle$
12.57 ps	13.50 ps	4.94 ps

where $\Delta\Phi$ is the angular width of the arc described by the Stokes vector on the Poincaré sphere, in the frequency window of width $\Delta\omega$. In order to reduce the uncertainty on $\Delta\tau$ due to error bars in the motion of the point on the Poincaré sphere, we adopted an interpolating procedure which takes into account groups of three points at a time, with a shifting scheme. As we said in the Introduction, we chose to use, as tunable source, a DFB temperature-tuned laser diode, emitting from 1549 to 1553 nm, and tuned it in steps of 0.1 nm. This gave us 41 points on the Poincaré sphere; a complete measurement was repeated every 30 min. Fig. 7 shows an example of the wavelength dependence of DGD, taken at a particular time. The DGD oscillates between a minimum of 8 ps and a maximum of 20 ps, with several cycles in the spectral range considered. This shows that the DGD is strongly dependent on wavelength, and can vary even by a factor of 2, in a wavelength range of less than 0.3 nm. Fig. 8 shows the time evolution of the DGD (averaged over a 4 nm wavelength interval). If the DGD is averaged both versus wavelength and versus time, we find a value of 12.57 ps. The PS method lends itself to further exploitation. We can derive experimentally the statistical distribution of DGD versus time or versus wavelength, which, according to theory, should be Maxwellian [2]. The wavelength and time dependence of DGD, and information about statistics, are summarized in Table I, which shows time-averaged values over 25 h.

To compare these data to the theoretical Maxwellian probability distribution, consider the ratio between the third and the first column. It gives a value (0.393) which is slightly smaller than the theoretical one ($\sqrt{(3\pi - 8)/8} \approx 0.422$). Similarly,

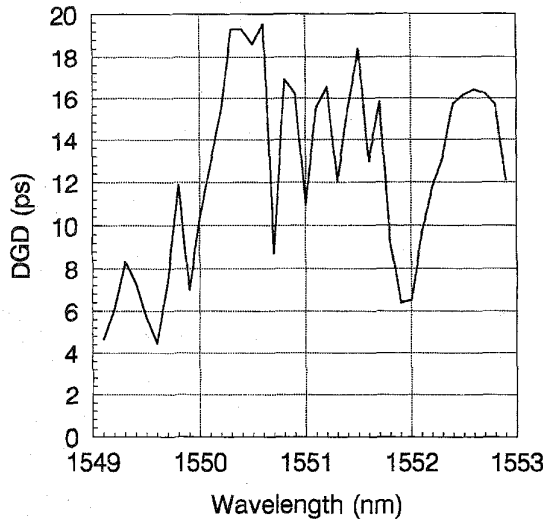


Fig. 7. DGD behavior as a function of wavelength for PS method at a fixed instant.

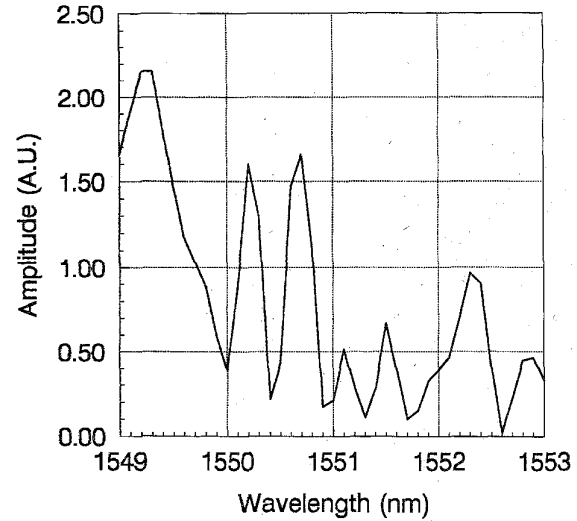


Fig. 9. Output curve using SOP method in the wavelength window 1549-1553.

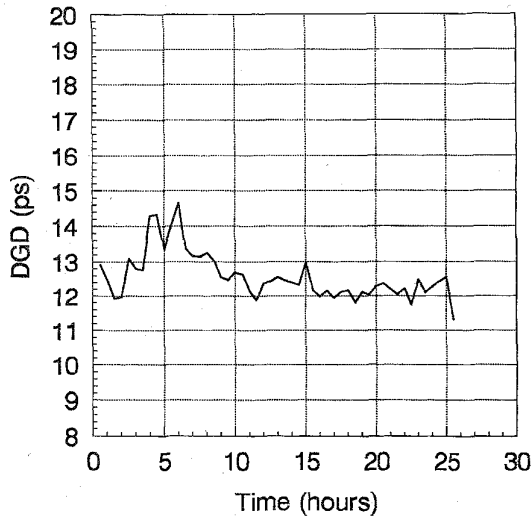


Fig. 8. Time evolution of DGD calculated by means of PS method; the wavelength range is 1549-1553 nm.

the ratio between the second and the first column is 1.074, again a little less than the theoretical value ($\sqrt{3\pi/8} \approx 1.085$). These comparisons indicate that the experimental statistical behavior is very close to the theoretical ones. The slight discrepancies might be due to the fact that the fibers, as we said, are in a "quasilong-length" regime, but the amount of available data does not allow us to draw, for the time being, a firm conclusion in this sense.

D. The State of Polarization Method

This method can be looked at as a simplified version of PS, since it allows to determine the average DGD over a given spectral range, which, in our case, was about 4 nm. The results to be presented in this sub-section were obtained

in the following way. The outputs of the polarimeter used for the PS method see Fig. 1, i.e., the Stokes parameters (S_0, S_1, S_2, S_3), are fed into a PC and processed by means of a package that simulates a polarization analyzer. The final result is a SOP curve as a function of wavelength. The average DGD is then calculated as follows:

$$\langle \Delta\tau \rangle = \pi \left(\frac{N_e}{\Delta\omega} \right) = \frac{N_e}{2c} \left(\frac{\lambda_1 \lambda_N}{\lambda_N - \lambda_1} \right) \quad (7)$$

where N_e is the number of extrema in the SOP curve between λ_1 and λ_N , i.e., in the frequency window of width $\Delta\omega$. The axes of the analyzer may be oriented at an arbitrary angle with respect to a fixed reference frame. Four different orientations ($0^\circ, 30^\circ, 45^\circ, 60^\circ$) were used. The four SOP curves are similar, qualitatively. A typical example is shown in Fig. 9. Unfortunately, the spectral range that our source could scan was small and, as a consequence, N_e changed from 10 to 16, depending on the analyzer orientation. We averaged the DGD's on 2 orientations ($0^\circ, 30^\circ$), 3 orientations ($0^\circ, 30^\circ, 60^\circ$) and four orientations ($0^\circ, 30^\circ, 45^\circ, 60^\circ$), in order to study how this affects the uncertainty on the calculated value of $\langle \Delta\tau \rangle$. The results are summarized in Fig. 10. While the curves averaged over 3 and 4 orientations are almost the same, they differ completely from the first one (average over 2 orientations). We conclude that three different angles for the analyzer are a good tradeoff between accuracy and measurement time. Fig. 11 shows the $\langle \Delta\tau \rangle$ evolution as a function of time. The averaged value (over time and over wavelength) was 13.67 ps, for 3 analyzer angles, and 13.74 for 4 angles. It is 8% larger than the results obtained with the PS method. Notice that this difference is not necessarily a sign of measurement errors; in fact, the wavelength intervals that were scanned for the SOP test and for the PS one, did not coincide. The first one was part of the second one, since λ_1 and λ_N correspond to two extrema in the SOP curve that fall inside the 1549-1553 nm range.

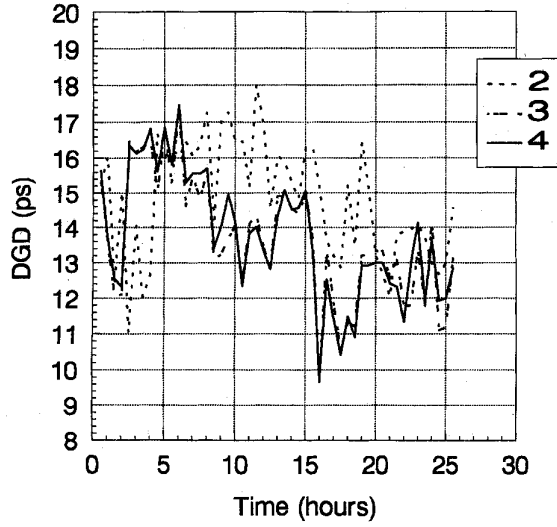


Fig. 10. Time evolution of DGD's as a function of number of different angles between analyzer and output state of polarization: 2 angles (dotted line), 3 angles (dashed-dotted line) and 4 angles (continuous line). The orientations were, respectively, 0° and 30° (dotted line); 0° , 30° , 60° (dashed-dotted line); 0° , 30° , 45° , 60° (continuous line).

IV. ESTIMATE OF THE CORRELATION LENGTH

At the present stage, all the methods seem to agree on the point that the link under test does not operate in what is currently referred to as the long-length regime, ($L/l_c \rightarrow \infty$). But none of them is capable of giving us an experimental figure for the correlation length. We may proceed to infer it as follows.

We measured the per-unit-length differential group delay on an ensemble of DS fibers, made by the same manufacturer as those in the field-tested cable, loosely set on laboratory benches. It turned out to be 0.4-0.6 ps/km. These values are of the same order of magnitude as those that were reported before for other DS fibers [2], [18]. We saw before that the DGD probability distribution is essentially Maxwellian; thus, the average DGD of the whole link can be expected to grow like the square root of $L \cdot l_c$. Then for a total DGD of the link around 13 ps, and a link length of 168 km, we estimate [2], [16] that the correlation length of the perturbation affecting the fiber is larger than 1 km. This result confirms that the ratio L/l_c is very large but finite. Even if we take a safety margin on the per-unit-length DGD of the unperturbed fiber, the conclusion remains, qualitatively, the same. Notice that per-unit-length differential group delays in standard step-index fibers are of the same order of magnitude as those that we measured on DS fibers. The much larger DGD's that were measured over long links made of DS fibers appear then to be caused by the correlation lengths, which seem to be more than one order of magnitude larger than in standard single-mode fibers.

V. CONCLUSIONS

Fig. 12 summarizes DGD versus time, as obtained with three techniques, over the same spectral range, 1549-1553 nm.

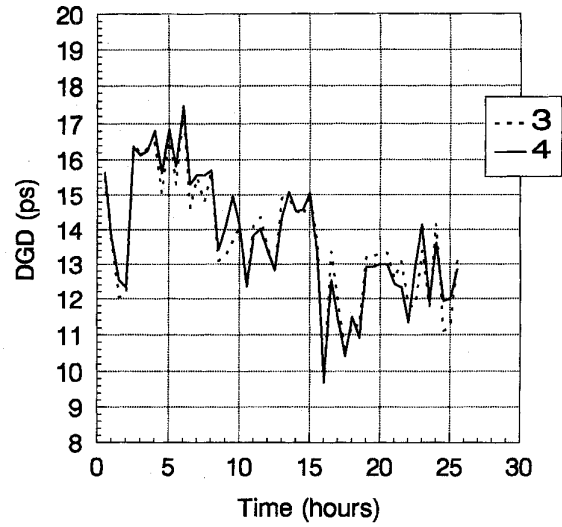


Fig. 11. Time evolution of DGD determined by means of SOP method using 3 different orientations (dotted line) and 4 different orientations (continuous line).

Error bars can be estimated to be of a few percent for PS (where errors are due mostly to interpolation of the motion on Poincaré sphere), slightly larger for the SOP method, and about 18% for the FP method. In spite of these differences in accuracy, average values given by the three techniques are almost equal (13 ± 0.5 ps). The IF method was not included in this comparison because it cannot be used over such a small spectral range. The $\langle\langle \Delta\tau \rangle\rangle$, i.e., the average over time and over 4 nm, is equal to 12.57 ps (for PS), 12.57 ps (for FP, with $k = 0.916$), 13.76 ps (for SOP). These three methods are then suitable for DGD field tests, even over very long links. However, let us make the following comments.

- The PS method is preferable, as it permits a complete characterization of PMD in a spectral range of arbitrary width; its drawback is that it needs a fast polarimeter, which is usually an expensive instrument.
- The SOP method is faster than the PS one, but it provides only an average value of the DGD, although on an arbitrary spectral range. The polarimeter can be slower than for PS. No information can be drawn about statistics.
- The FP method yields an average DGD, with a small error bar, only over a large spectral range (typically, at least a few THZ). It allows also to check in which regime (long- or short-length) the fiber under test is. In our opinion, only the spontaneous emission of an EDFA provides a suitable light source, since a high dynamic range is necessary.
- The IF method allows to determine only the average DGD over a very large spectral range, which can be quite different from the value at the wavelength of interest.

For what refers to the results of our in-field measurement campaign, it showed that dispersion-shifted (DS) fibers may

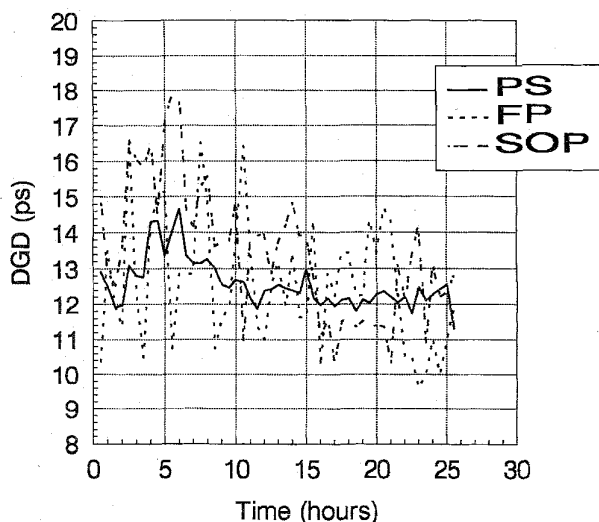


Fig. 12. Time evolution of DGD's obtained using 3 different methods over the same spectral range (1549-1553 nm): PS (continuous line), FP (dotted line), SOP (dashed-dotted line).

have a DGD larger than standard step-index fibers. This increase does not seem to be caused by a difference in per-unit-length birefringence of the fibers themselves, but to perturbation correlation lengths, which, in the case of DS fibers, appear not to be very small compared to the link length. In particular, in our experiment the perturbation correlation length was estimated to be in the range of one kilometer.

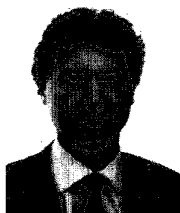
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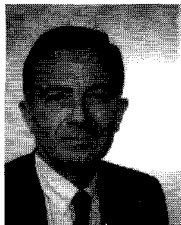
Andrea Galtarossa (M'88) was born in Padova, Italy, in 1958. He received the degree of electronic engineer from the University of Padova in 1984. During his last semester at this University, he received a thesis fellowship from Telettra spa, Italy.

In 1986, he received a postgraduate fellowship from Telettra for research on WDM components. In 1987, he joined Saifo, a private company involved in measurements and teaching on optical telecommunication systems. In 1990 he became an associate researcher in electromagnetic fields at the Department of Electronic and Informatics, University of Padova, Italy. He is author or co-author of more than 40 papers. His current research interests are in polarization properties of single-mode fibers, polarization mode dispersion measurements and optical devices modeling.

Dr. Galtarossa is a member of AEI.

Giovanni Gianello was born in Monza, Italy, on November 23, 1965. He received the degree in electronic engineering from the University of Padova in 1990.

From 1991 to 1992, he was engaged in research on optical cables splicing techniques. In 1994, he completed his doctoral studies at the University of Padova. His main research activity is on nonlinear optics and on optical devices modeling.



Carlo G. Smeda (S'63-M'66-SM'82) was born in Padova, Italy, in 1941. He received the E.E., M.S.E.E., and Libera Docenza degrees from the University of Padova, Italy, in 1964, Stanford University in 1966, and the Italian Ministry of Education in 1971, respectively.

While at Stanford, he was a recipient of the Volta Fellowship of the IEEE. Most of his research and teaching activities took place at the Universities of Trieste (1965-1967), Bologna (1968-1982), and Padova (since 1982). His main interests have been in the fields of nonlinear circuits, millimeter-wave devices, and optical waveguides. In 1972, he spent six months at Bell Laboratories, Holmdel, NJ, as a NATO exchange scientist working on early single-mode fibers. He became a full Professor in 1976, and served as Director of the Electronics Department at the University of Bologna, Italy, from 1978 to 1981. He is the author or co-author of about 110 papers, 3 textbooks, and several patents. His present interests are in linear and nonlinear optical guided-wave devices.

Dr. Smeda is a member of AEI and was Chairman of the AEI Special Group on Optoelectronics since its foundation in 1982 until 1986.



Marco Schiano (M'92) was born in Padova, Italy in 1964. He received the degree in electronic engineering from the University of Padova in 1990.

In 1991, he joined Saifo srl. (research and development of optical fibers for industrial applications), where he is now working in the fields of characterization of birefringent optical fibers and components and of optical instrumentation.